NO. 19 · VOL. FULL STACK    CURRICULUM VITÆ — THE FULL-STACK RECORD    EST. MMXIV

# Ahmad Abdul-Jawad *Mahmoud*

FULL-STACK ENGINEER / FOUNDER & CTO, BURAQ

CAIRO ⟶ RIYADH · DATABASE TO APP STORE, END TO END

CORRESPONDENCE
ahmad_abdulgawad.mahmoud@yahoo.com

TELEPHONE
+966 507 242 916 · +20 112 687 3152

BUREAU
Riyadh, Saudi Arabia

WIRE
LinkedIn — /ahmadabdulgawadmahmoud

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Full-stack engineer who owns a feature from the database to the app on the phone. Since March 2026 I am **Founder & CTO of Buraq** (buraqegypt.com), an Egyptian ride-hailing platform I built **solo, across every layer**: a NestJS / TypeScript backend on PostgreSQL and Redis — **22 modules, 330+ endpoints, 2,900+ tests** — a 29-screen Next.js operations console, the public web app live at buraqegypt.com, realtime Socket.IO feeds, payments, and native iOS & Android apps on one API contract. Behind it are **12+ years** of production engineering, seven of them owning the architecture of a POS used by 20k+ restaurants — so the APIs I design are the ones I would want to consume.

**9** REPOS BUILT BY ONE AUTHOR

**330+** API ENDPOINTS DESIGNED & SHIPPED

**2,900+** BACKEND TESTS I WROTE

**12+** YEARS IN PRODUCTION ENGINEERING

## § 01 Across the Stack

01 — BACKEND & DATA

### Correct before clever

A NestJS modular monolith (**22 modules**) on PostgreSQL with versioned migrations; Redis for dispatch state, the driver geo index, rate limits and cron leases. A **double-entry ledger**, a transactional outbox and idempotency keys, so money and side effects survive retries.

02 — REALTIME & INTEGRATIONS

### The parts that talk to the world

Socket.IO gateways for riders, drivers and admins; Google Maps routing with traffic-aware fare quotes and caching; Paymob cards, Apple Pay and Google Pay; WhatsApp-first OTP with SMS fallback; APNs and FCM push.

03 — WEB

### An ops console people can run a business on

A **29-screen** Next.js 14 console: live MapLibre trips map, dispatch and SOS boards, KYC review, finance reconciliation, refunds and payouts, a map zone editor, feature flags, **RBAC** and an audit log behind Google SSO.

04 — MOBILE CLIENTS

### 12 years of shipping apps

Native SwiftUI (Swift 6) and Kotlin / Compose apps on shared, versioned core libraries, with **1,000+ iOS** and **4,000+ Android tests**. Before this: apps used by **35M+ people** — HungerStation, Ego ride-hailing and the Marn POS.

## § 02 Professional Experience

Mar 2026 — Present
CURRENT
CAIRO · FOUNDER

### Founder & CTO — Backend · Web · iOS · Android

Buraq — buraqegypt.com ↗ · Buraq Technologies

Founded an Egyptian ride-hailing platform and have carried its **entire technical scope alone**: chose the stack, set the standards, and built eleven repositories from empty — a hardened backend and operations console, rider and captain apps on **iOS and Android**, a shared core package per platform, and the public web app **live at buraqegypt.com**. Every layer governed by explicit rules: idempotent mutations, double-entry accounting, acyclic module boundaries, and a CI suite that gates the merge. Now preparing a cash-only lean beta in Cairo.

**Backend API**
NestJS · TypeScript · Mar '26 → now

Modular monolith, **22 modules** in an acyclic graph — **trip state machine**, **scored waterfall dispatch** over a Redis geo index, admin-tunable pricing across six ride tiers, **double-entry ledger** with cash and card settlement (Paymob, Fawry), scheduled and multi-stop rides, five realtime **Socket.IO gateways**, OTP / Apple / Google auth with RBAC, idempotency and audit log · **330+ endpoints** over 55 controllers, 62 entities, **2,900+ tests**, Swagger docs.

**Ops Console**
Next.js 14 · Jul '26 → now

**29-screen** ops surface — live **MapLibre** trips map with demand heatmap, dispatch and SOS boards, KYC review, finance reconciliation, refunds and payouts, cash control, map **zone editor**, feature flags, audit log, **Google SSO** · English and Arabic, 170+ tests.

**iOS · Rider & Captain**
SwiftUI · Swift 6 · Jul '26 → now

Two native **SwiftUI** apps on a shared **BuraqCore** package (domain · networking · design system, version-pinned) — the full **trip lifecycle** on live GPS, Socket.IO feeds with fresh-token reconnect, chat, wallet and payments, **Live Activities**, Siri voice booking, on-device ID verification, captain earnings and navigation · ~240 views, **1,000+ tests**, strict concurrency.

**Android · Captain**
Kotlin · Jetpack Compose · Aug '26 → now

Native port against **the same API contract** — 18 feature modules on convention plugins, Hilt, a pure-Kotlin domain; onboarding, the shift's foreground service and the trip end to end running against the backend · **1,500+ tests**.

**Android · Rider**
Kotlin · Jetpack Compose · Sep '26 → now

Screen-by-screen parity port of the iOS rider app — 11 feature modules on a published **buraq-core-android** (GitHub Packages) · **2,200+ tests**, including 371 screenshot goldens in light, dark and RTL.

**Web · buraqegypt.com**
Next.js 16 · React 19 · Oct '26 → now

Marketing site, **rider web app** with live trip tracking, **captain onboarding portal** with on-device document OCR, public trip-share pages and app links · English and Arabic (RTL), WCAG 2.2 AA, **1,800+ tests** · **live on Render since 15 Oct 2026**.

**iOS · Buraq Schools**
SwiftUI · Swift 6 · Nov '26 → now

A second product on the same platform: **one app for parents and school-bus supervisors** — the live bus, safe handovers, an offline trip mode and an **iOS 26 Liquid Glass** interface · **1,000+ tests**; the Android port is under way.

One backend · one contract · eight client surfaces — and the company around them: incorporation, the Law 87/2018 licence path, a pricing engine fitted to 626 observed fares, the business model and investor materials, the go-live plan. 3,100+ commits, one author — so I know exactly what I will be asking of the team I hire.

Aug 2019 — Aug 2026
RIYADH, KSA

### iOS Team Lead — 7 years leading the iOS org

Marn — Jahez Group ↗

- **Led a 5+ engineer iOS team** for seven years — onboarding, mentoring, career growth, and the review culture behind an enterprise POS used by 20k+ restaurants and cafés.
- **Owned the technical roadmap and architecture:** an offline-first CoreData core with cloud sync, built so a venue keeps trading when the network does not.
- **Set the engineering standards** — module boundaries, code review, automated testing — that held the app at a 99.8–99.9% crash-free rate in production.
- **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery, and made the case for it to product leadership.
- **Built CI/CD and the release process from scratch**, cutting deployment time 60% and shortening every engineer's feedback loop.
- Ran the performance programme (40% faster load times) and designed the table-management, order-processing and inventory systems at the heart of the product.
- Partnered with product, design and backend leadership on scope, estimates and delivery commitments the team could keep.

Dec 2025 — Jun 2026
VOLUNTEER

### Senior iOS Engineer

HaramBlur ↗ · Voluntary Charity Project

- Designed an architecture integrating on-device machine learning for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implemented efficient image processing with region-based and full-image blur effects.
- Created a real-time video monitoring system for continuous content filtering.
- Developed smart banner blocking and optimised memory management throughout.

Mar 2018 — May 2019
RIYADH, KSA

### iOS Team Lead

Ego — The Ride Hailing App ↗

- Led iOS delivery for a complete ride-hailing ecosystem (rider, driver, watchOS) serving 15M+ users in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone ⟷ Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

Sep 2016 — Mar 2018
KSA

### Senior iOS Engineer

HungerStation — DeliveryHero Group ↗

- Contributed to a major food-delivery platform serving 20M+ users.
- Optimised app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery optimisation.
- Collaborated with backend teams on API design and integration.

May 2016 — Aug 2016

### iOS Developer

ALNapedh ↗

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Spreedly payments and live-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

Jul 2015 — Sep 2015

### iOS & Apple Watch Developer

Nile Design Co. ↗

- Early adopter of Apple Watch development (watchOS 1–2).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from WWDC 2019 onward.

May 2014 — May 2015

### iOS Developer

UK Pro. Solutions Ltd. ↗

- Foundation iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

## § 03 Selected Projects

01 — FULL-STACK

### Buraq — Ride-Hailing (founded Mar 2026)

NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js · SwiftUI · Kotlin · Jetpack Compose

Egyptian ride-hailing platform I founded and built solo: a hardened NestJS backend (trip state machine, scored dispatch, double-entry ledger, five realtime gateways, 2,900+ tests), a 29-screen ops console, rider and captain apps on **iOS and Android**, and the public web app **live at buraqegypt.com**.

◆ Founder & CTO · Nine Repos · 10k+ Tests

02 — ENTERPRISE

### Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise point-of-sale serving 20k+ restaurants and cafés across Saudi Arabia. Offline-first architecture, complex table management, real-time order processing — **the product my iOS team owned for seven years**.

★ 4.8 / 5 — 2,437 ratings

03 — MOBILITY

### Ego — Ride Hailing

MapKit · Core Location · Apple Pay · Push · watchOS

Complete ride-hailing ecosystem: rider app, driver app, and Apple Watch companion. Real-time GPS, Apple Pay, and cross-device experiences for 15M+ users.

★ 4.7 / 5 — 211.8k+ ratings

04 — DELIVERY

### HungerStation

UIKit · Firebase · Real-time APIs · Perf Optimization

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Focused on performance and real-time order tracking.

★ 4.5 / 5 — 1.9M+ ratings

05 — AI / PRIVACY

### HaramBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. On-device ML, smart blur effects, and live video monitoring.

◆ AI-Powered Privacy Browser

06 — COMMUNITY

### Daleelo

MapKit · Core Location · Social SDKs · Community

Muslim community app for users across the US — prayer times, mosque finder, and community features with enhanced location and social integration.

★ 4.5 / 5 — 1,287 ratings

## § 04 Stack & Craft

WEB & INFRASTRUCTURE — FULL-STACK

`Next.js 14` `React` `TypeScript` `MapLibre` `Socket.IO` `Docker` `GitHub Actions CI` `Prometheus Metrics` `Jest` `Google Maps Platform` `Paymob` `Cloudflare R2`

IOS — 2025 — CORE

`Swift 6.0` `SwiftUI` `UIKit` `Combine` `async/await` `SwiftData` `iOS 18+`

ANDROID — 2025 — NATIVE

`Kotlin` `Jetpack Compose` `Coroutines` `Flow` `MVVM`

ARCHITECTURE — PATTERNS

`MVVM` `VIPER` `Clean Architecture` `Coordinator` `TDD`

APPLE ECOSYSTEM — PLATFORMS

`watchOS` `tvOS` `macOS` `WidgetKit` `Live Activities` `App Intents`

MODERN FEATURES — INTELLIGENCE

`Core ML` `Vision` `ARKit` `Dynamic Island` `App Tracking`

BACKEND & APIS — CORE

`NestJS` `TypeScript` `PostgreSQL` `TypeORM` `Redis` `JWT/OTP Auth` `REST` `GraphQL` `Firebase` `CloudKit` `Socket.IO` `Swagger/OpenAPI`

TOOLING & AI — WORKFLOW

`Xcode 15+` `Fastlane` `TestFlight` `Instruments` `Git` `Claude Code` `Copilot` `Cursor`

## § 05 Portfolio Index

FOOD & GAMING
HungerStation — delivery
Punters — racing tips

TRANSPORTATION
Buraq — founder
Ego — Rider
Ego — Driver
Ego — Watch app

ENTERPRISE
Marn POS
Medical Guide

SOCIAL & COMMUNITY
Daleelo
NU-inc
VeeU

WATCHOS
Ego Watch
TaQs Weather
TweetMe

E-COMMERCE
iBss
Exclusive Luxe
Scratch

RELIGIOUS & EDU
Afasy
ArabNileGroup

UTILITY & TOOLS
Unlocking Smart
Insta
iContacts
🚀 *First iOS app*

AI & ML
HaramBlur — CoreML
RecognizeMe
🎓 *Grad project — OpenCV*

WEB
FaceDemo
Web Pages

## § 06 Record

EDUCATION

**B.Sc. Information Systems**
Computer Sciences College
Asyut University, Egypt · 2009–2013
iOS Semester — Excellent (91.5%)

CERTIFICATION

**ISTQB Foundation Level**
Egyptian Software Testing Qualification Board

LANGUAGES

| | |
|---|---|
| Arabic | Native |
| English | Professional |

## § 07 Honours

01 Building **Buraq solo across every layer** — backend, ops console, web app, and iOS and Android apps on one API contract.

02 Held a **99.8–99.9% crash-free rate** and cut deployment time **60%** by building the standards, tests and CI/CD the team ran on.

03 **Founded Buraq** (buraqegypt.com), an Egyptian ride-hailing platform, and built all eleven repositories solo — backend, ops console, rider and captain apps on iOS and Android, Buraq Schools, and the public web app, live since Oct 2026 — behind **10,000+ automated tests**.

04 Built a food-delivery platform serving **20M+ users** across Saudi Arabia and Bahrain, and a ride-hailing ecosystem serving **15M+ users** in Saudi Arabia.

05 **10+ apps** published on the App Store, used by millions across MENA.

06 **30+ projects** delivered across a 12-year career.

07 Early Swift adopter (WWDC 2014) and early SwiftUI adopter (WWDC 2019) — a habit of betting on the right platform shift early.

FULL-STACK ENGINEER · FOUNDER & CTO, BURAQ · BACKEND · WEB · MOBILE · 12+ YEARS

SET IN FRAUNCES & JETBRAINS MONO    EDITION 2026 · CAIRO ⟶ RIYADH